AMC 8 Style Practice Test

25 problems · 40 minutes · no calculator · choose one answer (A–E) for each problem

Name: ______________________ Date: ______________________

1. Maya has 3 packs of 12 stickers each and 7 loose stickers. How many stickers does Maya have in all?

(A) 36 **(B)** 40 **(C)** 43 **(D)** 46 **(E)** 55

2. Which of the following fractions is the largest?

(A) $\frac{2}{5}$ **(B)** $\frac{3}{7}$ **(C)** $\frac{4}{9}$ **(D)** $\frac{1}{2}$ **(E)** $\frac{5}{9}$

3. The perimeter of a rectangle with side lengths 3 and 7 is

(A) 10 **(B)** 20 **(C)** 21 **(D)** 24 **(E)** 42

4. 100 candies are packed into bags of 7 candies each. How many candies are left over?

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** 5

5. What is 25% of 84?

(A) 19 **(B)** 20 **(C)** 21 **(D)** 25 **(E)** 28

6. A movie starts at 2:45 pm and runs for 1 hour 50 minutes. When does it end?

(A) 4:25 pm **(B)** 4:30 pm **(C)** 4:35 pm **(D)** 4:45 pm **(E)** 4:55 pm

7. A cyclist rode 3 miles in 30 minutes. What was the cyclist’s speed in miles per hour?

(A) 1.5 **(B)** 3 **(C)** 4.5 **(D)** 6 **(E)** 9

8. The number 40 is split into two parts in the ratio 2 : 3. What is the larger part?

(A) 16 **(B)** 18 **(C)** 20 **(D)** 24 **(E)** 28

9. In the sequence 7, 11, 15, 19, ... each term is 4 more than the previous one. What is the tenth term?

(A) 43 **(B)** 45 **(C)** 47 **(D)** 49 **(E)** 51

10. The numbers on all six faces of a die add up to 21. The die sits on a table; an observer sees three of its faces, showing 1, 2, and 3. What is the sum of the numbers on the three hidden faces?

(A) 12 **(B)** 13 **(C)** 14 **(D)** 15 **(E)** 16

11. A toy costs $90. During a sale, its price is reduced by 30%. What does the toy cost now (in dollars)?

(A) 27 **(B)** 57 **(C)** 60 **(D)** 63 **(E)** 70

12. A square of side 2 is cut from the corner of a square of side 6. What is the area of the remaining figure?

(A) 24 **(B)** 26 **(C)** 28 **(D)** 30 **(E)** 32

13. How many two-digit numbers have digit sum 9?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

14. Bus 1 leaves a stop every 12 minutes, and bus 2 every 18. They leave together at noon. In how many minutes will they next leave together?

(A) 24 **(B)** 30 **(C)** 36 **(D)** 54 **(E)** 72

15. A bag holds 3 red and 5 blue marbles. One marble is drawn at random. What is the probability that it is red?

(A) $\frac{1}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{3}{8}$ **(E)** $\frac{1}{2}$

16. At a party, each of 8 people shook hands with every other person exactly once. How many handshakes were there?

(A) 16 **(B)** 28 **(C)** 36 **(D)** 56 **(E)** 64

17. A father is 38 and his son is 8. In how many years will the father be exactly twice as old as his son?

(A) 11 **(B)** 15 **(C)** 18 **(D)** 22 **(E)** 30

18. A third of a class left for a field trip, then a quarter of those remaining left for a rehearsal. Eighteen students stayed in the room. How many students are in the class?

(A) 24 **(B)** 27 **(C)** 32 **(D)** 36 **(E)** 48

19. How many times does the digit 7 appear when the numbers from 1 to 100 are written out?

(A) 10 **(B)** 16 **(C)** 18 **(D)** 19 **(E)** 20

20. How many three-digit numbers have three different digits, all of them odd?

(A) 45 **(B)** 48 **(C)** 60 **(D)** 75 **(E)** 100

21. What is the least positive integer greater than 1 that leaves a remainder of 1 when divided by each of 2, 3, 4, and 5?

(A) 21 **(B)** 31 **(C)** 41 **(D)** 61 **(E)** 121

22. A two-digit number increases by exactly 75% when its digits are reversed. How many such two-digit numbers are there?

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 8

23. A rectangle has perimeter 26 and area 40. By how much is the longer side greater than the shorter side?

(A) 2 **(B)** 3 **(C)** 5 **(D)** 8 **(E)** 13

24. Anya has 15 coins, nickels and quarters, worth $2.15 in all. How many quarters does she have?

(A) 5 **(B)** 6 **(C)** 7 **(D)** 8 **(E)** 9

25. In how many zeros does the product $1 \cdot 2 \cdot 3 \cdot \ldots \cdot 25$ end?

(A) 5 **(B)** 6 **(C)** 7 **(D)** 8 **(E)** 10

Answer Key and Solutions

Parent page: cut off or do not print for the student. Solutions are on the test page online.

1	2	3	4	5
C	**E**	**B**	**B**	**C**
6	7	8	9	10
C	**D**	**D**	**A**	**D**
11	12	13	14	15
D	**E**	**B**	**C**	**D**
16	17	18	19	20
B	**D**	**D**	**E**	**C**
21	22	23	24	25
D	**B**	**B**	**C**	**B**

1. $3 \cdot 12 + 7 = 43$.

2. Compare with one half: only 1/2 and 5/9 are at least a half, and $5/9 > 1/2$ (since $5 \cdot 2 > 9$).

3. $2(3 + 7) = 20$.

4. $100 = 14 \cdot 7 + 2$: remainder 2.

5. A quarter of 84: $84/4 = 21$.

6. 1 hour 50 minutes after 2:45 pm is 4:35 pm.

7. 3 miles in half an hour is 6 miles per hour.

8. One part = $40/5 = 8$: the parts are 16 and 24; the larger is 24.

9. $7 + 9 \cdot 4 = 43$ (nine steps of 4).

10. $21 - (1 + 2 + 3) = 15$.

11. 30% of 90 = 27, so $90 - 27 = 63$.

12. $36 - 4 = 32$.

13. The tens digit runs 1 through 9 and determines the units digit: 18, 27, ..., 90 — nine numbers.

14. LCM(12, 18) = 36.

15. 3 red out of 8 marbles: 3/8.

16. Every pair of 8 people: $8 \cdot 7/2 = 28$.

17. $38 + x = 2(8 + x)$: $x = 22$.

18. $(2/3) \cdot (3/4) = 1/2$ of the class stayed, so the class has 36 students.

19. Ten times in the units place (7, 17, ..., 97) and ten in the tens place (70–79): 20 in all.

20. There are five odd digits (1, 3, 5, 7, 9): $5 \cdot 4 \cdot 3 = 60$.

21. One more than a multiple of LCM(2,3,4,5) = 60: that is 61.

22. If the number is $10a+b$, then $10b+a = 1.75(10a+b)$, so $b = 2a$: the numbers are 12, 24, 36, 48 — four of them.

23. The sides add to 13 and multiply to 40: they are 5 and 8, difference 3.

24. With q quarters, $25q + 5(15 - q) = 215$: $q = 7$.

25. Each zero needs a 2×5 pair; fives are scarcer: they come from 5, 10, 15, 20, and 25 (which gives two), six in all.